

SUVRADIP GHOSH

West Bengal, India | suvrawork03@gmail.com | +91 8653056408 | LinkedIn | GitHub | Portfolio

EDUCATION

Master of Computer Applications	2024 – 2026
Sikkim Manipal Institute of Technology	CGPA: 8.69/10
Bachelor of Computer Applications	2021 – 2024
Techno India University	CGPA: 8.39/10

TECHNICAL SKILLS

Programming Languages	Python, Java, JavaScript, C
Frontend	React.js, HTML, CSS, Tailwind CSS, Redux, Zustand
Backend	Node.js, Express.js, FastAPI
Databases	MongoDB, SQLite, Redis
AI / ML	PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib
Tools	Git, Docker, Postman, Jest, Pytest

EXPERIENCE

Android Developer Intern - EduSkills Remote	Apr 2024 – Jun 2024
- Built Android applications with Kotlin and Android Studio, implementing UI and logic for assigned tasks. - Debugged and tested applications, fixing implementation issues during assignments and technical assessments.	

PROJECTS

InsightATS Hiring Platform with Project & Context Based Screening	Live GitHub
Tech Stack: React.js • Redux • FastAPI • PyTorch • Redis • BERT • RoBERTa • DistilBERT • SHAP	

- Developed a hiring platform that scores candidates on **Project Complexity** and **Contextual Resume-JD** matching instead of only keyword dependence.
- Reduced inference latency 51%** (6.0s → 3.0s) by lazy-loading models in FastAPI and caching results in Redis.
- Batched resume scoring against a shared job description in a single request instead of one call per resume, thereby **cutting client-to-server API requests by up to 90%**.
- Fine-tuned BERT, RoBERTa, and DistilBERT for resume parsing, resume-JD matching, and project classification, achieving **0.75 F1** and **0.78–0.93 Macro F1** across the three tasks.

CodeNest Cloud IDE with Containerized Execution & AI Features	Live GitHub
Tech Stack: React.js • Zustand • Express.js • Docker • MongoDB Atlas • Redis • Groq Llama-3.3-70B	

- Built a cloud IDE running code in **isolated Docker containers** (5s timeouts), supporting **20+ concurrent users** with zero cross-contamination.
- Queued code-execution requests through a **25-job async pipeline** with dedicated workers, maintaining system stability during concurrent load spikes and preventing resource exhaustion.
- Integrated **Groq Llama-3.3-70B** for code review, debugging, and flowchart generation - **cut response latency to 1–3 seconds** via Redis caching and an in-memory fallback.
- Replaced nested React Context with persistent **Zustand** stores, eliminating prop-drilling across auth, project, and version state.

Skin Disease Detection using CNN	Kaggle GitHub
Tech Stack: Python • TensorFlow • EfficientNetB3 • OpenCV • Scikit-Learn • CNN	

- Fine-tuned **EfficientNetB3** on **10,015 dermatoscopic images** from the **HAM10000** dataset, achieving **90% test accuracy** for skin disease classification.
- Prevented data leakage and improved generalization via **patient-wise train/test splits** and **CLAHE** contrast enhancement with hair-artifact removal.
- Addressed class imbalance using focal loss and test-time augmentation, lifting per-class F1 to **0.94 (NonMelanoma)**, **0.89 (Benign)**, and **0.86 (Melanoma)**.

CERTIFICATIONS

IBM Professional Certificate: Full Stack Software Developer	Credential
IBM Professional Certificate: Data Science	Credential